👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.AT.7

Identify Equivalent Algebraic Expressions

Lesson 6-6 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can show that two expressions are equivalent by simplifying and combining like terms.

Language Objective: I can explain my work using the words equivalent, simplify, like terms, and combine.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Equivalent Expressions · **Equivalent** · **Simplify** · **Like Terms** · **Combine** · **Coefficient** · **Commutative Property** · **Associative Property** · **Counterexample**

1. Expressions with the same value for _____ allowed number are equivalent.

2. Expressions that always have the same value are _____.

3. Rewriting an expression in a shorter form that is still equal to it is to _____ it.

4. Terms with the same variable raised to the same power, like 4x and 9x, are _____.

5. To add or subtract like terms into one term is to _____ them.

6. A number multiplied by a variable is a _____.

7. The rule that a + b = b + a is the _____.

8. The rule that (a + b) + c = a + (b + c) is the _____.

9. One value that shows two expressions are not equal is a _____.

Write About the Math

Are these expressions equivalent? How can you prove it?
¿Estas expresiones son equivalentes? ¿Cómo lo puedes probar?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Equivalent Expressions** *Expresiones equivalentes*
- ☐ **Equivalent** *Equivalente*
- ☐ **Simplify** *Simplificar*
- ☐ **Like Terms** *Términos semejantes*
- ☐ **Combine** *Combinar*

Model: *3x + 2x combines to 5x, so the two forms are equivalent.* — Students explain 3x + 2x = 5x (combining like terms), so both equal 5x + 10, and verify by substituting a value like x = 4 to get 30 both ways.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The expressions ___ equivalent because when I simplify 4h + 2h + 30, I get ___. I can prove it by substituting h = ___: both expressions equal ___.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Equivalent Expressions
2. **Notes 2:** Equivalent
3. **Notes 3:** Simplify
4. **Notes 4:** Like Terms
5. **Notes 5:** Combine
6. **Notes 6:** Coefficient
7. **Notes 7:** Commutative Property

8. **Notes 8:** Associative Property
9. **Notes 9:** Counterexample
10. **Watch for this mistake:** *A common mistake in Equivalent Expressions is adding the exponents instead of the coefficients when combining like terms — writing $3x + 2x$ as $5x^2$ instead of $5x$. Remember: $3x$ means 3 groups of x and $2x$ means 2 more groups of x, so you add the coefficients ($3 + 2 = 5$) and keep the variable's power the same (x^1, not x^2).*
11. **Try It 1:** C. $4x$ — *$3x + x$ is 3 groups of x plus 1 more group of $x = 4x$. Add the coefficients: $3 + 1 = 4$. Check $x = 3$: $3(3) + 3 = 12$ and $4(3) = 12$.*
12. **Try It 2:** A. $2x + 5$ — *Order of addition does not change the value (commutative property): $5 + 2x = 2x + 5$. Check $x = 10$: $5 + 2(10) = 25$ and $2(10) + 5 = 25$.*